Retirement Closure Control Module (RCCLM)

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OOF® Origin Open Foundation®

Independent Methodological Authority

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Category: Governance & Enforcement

Subcategory: Software Retirement Governance

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Canonical Definition

Retirement Closure Control Module (RCCLM) is the Core Module within Software Retirement Standard (SRTS) that governs how the actual outcomes of software retirement are verified, reconciled with the intended retirement state, evaluated against applicable closure requirements, and formally assigned a justified retirement disposition.

RCCLM establishes software retirement closure as an evidence-supported

governance determination rather than an administrative declaration that software has been discontinued, deleted, replaced, or removed from production.

The module distinguishes retirement execution from retirement verification,

migration completion from retirement completion, operational withdrawal from residual obligation resolution, and administrative closure from legitimate retirement finality.

It requires the closure assessment to preserve the identity of the retiring

software, the authorized retirement scope, the intended retirement state, the actual operational condition, support status, migration outcomes, residual dependencies, data relationships, authority conditions, verification evidence, unresolved obligations, and applicable closure authority.

RCCLM recognizes that retirement may be complete for a defined software version, environment, component, or capability while other versions, environments, or obligations remain active.

It also recognizes that legitimate retirement closure may coexist with continuing historical, evidentiary, contractual, data, or recovery obligations, provided those obligations are explicitly identified, assigned, and governed.

The module establishes the conditions under which a retirement case may be closed, conditionally closed, partially closed, deferred, rejected, or reopened.

RCCLM does not require every historical relationship to disappear. It requires the intended operational withdrawal to be verified and every material continuing obligation to have an appropriate governance disposition.

Its principal governance output is the Software Retirement Closure Profile .

Operational Role

RCCLM establishes the final verification and disposition mechanism within SRTS.

It governs:

• Retirement closure case identity

• Retirement scope reconciliation

• Intended retirement state verification

• Actual software state assessment

• Retirement execution evidence

• Software withdrawal verification

• Support status reconciliation

• Migration outcome reconciliation

• Residual dependency reconciliation

• Data continuity verification

• Interface withdrawal verification

• Access and authority reconciliation

• Operational responsibility continuity

• Automated execution termination

• Autonomous agent retirement conditions

• Recovery and historical retention obligations

• Retirement deviation assessment

• Retirement completeness

• Retirement evidence sufficiency

• Retirement uncertainty

• Closure eligibility

• Partial retirement closure

• Conditional closure

• Closure authority

• Closure decision

• Closure record integrity

• Continuing obligations

• Post-closure monitoring

• Closure invalidation

• Retirement case reopening

• Final retirement disposition

RCCLM does not replace the specialized mechanisms responsible for performing retirement activities.

It evaluates their outcomes for the purpose of establishing legitimate

retirement closure.

Operational Question

How can software retirement be formally closed only when the intended operational withdrawal has been verified, the actual retirement state has been reconciled with the approved scope, and every material remaining dependency, authority relationship, data obligation, and continuing responsibility has received an appropriate governance disposition?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting: Software Lifecycle Control

SRTS Core Module Flow

Retirement Planning Module (RPM) → End-of-Support Management Module (ESMM)

→ Software Migration Module (SMM) → Residual Dependency Management Module (RDMM) → Retirement Closure Control Module (RCCLM)

RCCLM is the fifth and final Core Module within SRTS.

It consolidates the governance outcomes of retirement planning, support

transition, migration, and residual dependency management.

Its purpose is not to repeat the activities of those modules but to determine whether their combined results establish the intended retirement state.

RCCLM also provides the verified retirement disposition to the

cross-cutting lifecycle governance mechanisms of SLGA™.

• Governed Space

Software Retirement Closure Governance

This governed space concerns the verification, reconciliation,

authorization, recording, and continuing validity of software retirement closure.

It begins when a retirement case becomes eligible for closure assessment or when a closure determination must be reconsidered.

It includes verification of the actual software state, comparison with the

intended retirement state, evaluation of material evidence, reconciliation of continuing obligations, and establishment of a justified closure disposition.

It remains applicable where retirement is incomplete, conditionally closed, partially closed, subject to material uncertainty, or reopened because of new evidence.

It concludes for a defined retirement case when an authorized closure disposition has been established and any continuing obligations remain appropriately governed.

Governed Object

The primary Governed Object is:

A Software Retirement Closure Case representing the evidence-supported

relationship between an identifiable retiring software object, its authorized retirement scope, intended retirement state, actual operational state, completed retirement activities, verified withdrawal conditions, residual dependencies, continuing obligations, applicable closure criteria, competent authority, and final retirement disposition.

The governed object must preserve, where applicable:

Retirement Closure Case ID

Retirement Case Reference

Software Object ID

Software Version and Configuration

Authorized Retirement Scope

Intended Retirement State

Actual Software State

Retirement Execution References

Support Status

Migration Outcome

Residual Dependency Disposition

Data Continuity State

Interface Withdrawal State

Authority and Access State

Verification Evidence

Unresolved Obligations

Closure Criteria

Closure Authority

Closure Decision

Closure Status

Closure Effective Time

Record Version

Scope

RCCLM governs:

Retirement Closure Initiation

Retirement Case Identification

Retirement Scope Verification

Intended–Actual State Reconciliation

Operational Withdrawal Verification

Support Condition Reconciliation

Migration Outcome Reconciliation

Residual Dependency Reconciliation

Data Obligation Reconciliation

Interface Withdrawal Assessment

Authority and Access Reconciliation

Autonomous Execution Reconciliation

Retirement Evidence Assessment

Retirement Deviation Assessment

Retirement Completeness Assessment

Closure Eligibility

Closure Authorization

Closure Classification

Conditional Closure

Partial Closure

Continuing Obligation Governance

Closure Record Preservation

Post-Closure Monitoring

Closure Reassessment

Retirement Case Reopening

RCCLM does not independently execute software deletion, migration, credential revocation, dependency replacement, or data disposal.

Foundational Principle

Software retirement is complete only when its intended operational withdrawal has been verified and its remaining dependencies, data relationships, authority conditions, and obligations have been appropriately resolved, transferred, or explicitly retained under governance.

A retirement announcement is not retirement evidence.

A successful migration does not establish source software withdrawal.

A software shutdown does not establish termination of every execution path.

And the absence of unresolved technical tasks does not automatically establish the absence of continuing obligations.

Retirement Declared ≠ Retirement Verified.

Administrative Closure ≠ Operational Finality.

Retirement Closure Case Identity

Every retirement closure determination must have a stable identity.

Relevant elements include:

Closure Case ID

Retirement Case Reference

Software Object Reference

Applicable Scope

Closure Assessment Origin

Responsible Function

Assessment Time

Closure Status

Evidence Reference

Record Version

The identity must remain traceable across closure decisions, reassessments, invalidations, and reopening events.

Closure Assessment Origin

Closure assessment may originate from:

Completion of Retirement Activities

Software Withdrawal Report

Migration Completion Report

Residual Dependency Resolution

Retirement Milestone

Support Termination

Operational Decommissioning

External Verification Requirement

Closure Reassessment

Previously Closed Case Reopening

The origin of a closure assessment does not establish that closure requirements have been satisfied.

Retirement Scope Reconciliation

RCCLM must establish the scope against which retirement closure is evaluated.

Relevant elements include:

Software Product

Software Version

Software Component

Deployment Environment

Operational Capability

Interface

User Group

Geographic or Organizational Scope

Associated Dependencies

Excluded Objects

Closure must not extend beyond the scope supported by evidence and applicable authority.

Intended Retirement State

RCCLM must retrieve or establish the intended retirement state defined through applicable retirement governance.

Potential intended conditions include:

Software No Longer Executing

Software No Longer Deployed

Service Withdrawn

Interface Terminated

Operational Capability Replaced

Users Migrated

Data Relationships Transferred

Dependencies Resolved

Permissions Revoked

Historical Artifact Retained Without Operational Use

Continuing Obligations Assigned

The intended state must remain sufficiently precise to support verification.

Actual Retirement State

RCCLM must establish the software's actual condition at closure assessment.

Relevant elements include:

Observed Execution State

Deployment State

Operational Availability

Interface Activity

User Access

Data Accessibility

Dependency State

Scheduled Execution

Service Credentials

Support Condition

Recovery Relationships

Known Operational Restrictions

The actual state must not be inferred solely from an administrative inventory label.

Intended–Actual State Reconciliation

RCCLM must compare the intended retirement state with the actual observed state.

Relevant elements include:

Intended Withdrawal Condition

Observed Withdrawal Condition

Intended Migration Outcome

Verified Migration Outcome

Intended Dependency Disposition

Actual Dependency Disposition

Intended Authority State

Actual Authority State

Intended Data Continuity

Actual Data Continuity

Material differences must remain explicit.

Retirement Execution Evidence

RCCLM must identify evidence of material retirement activities.

Potential evidence includes:

Decommissioning Records

Deployment Removal Records

Service Termination Records

Migration Execution Records

Access Revocation Records

Interface Withdrawal Records

Configuration Changes

Infrastructure Changes

Dependency Resolution Records

Operational Monitoring

Execution evidence must remain distinguishable from outcome verification.

Operational Withdrawal Verification

RCCLM must verify whether the software has been withdrawn from the intended operational scope.

Relevant elements include:

Software Identity

Deployment Environment

Execution Paths

Active Services

Scheduled Processes

User Access

External Interfaces

Automated Invocations

Operational Monitoring

Withdrawal Evidence

Verification must account for material execution paths beyond the primary application process.

Software Execution State

RCCLM must establish the relevant execution state.

Potential states include:

Active

Restricted

Partially Deactivated

Stopped

Disabled but Re-activatable

Removed From Deployment

Retained for Recovery

Verified Non-Operational

Unknown

A stopped process does not establish that the software cannot execute through another mechanism.

Deployment State Reconciliation

RCCLM must identify whether software remains deployed.

Relevant elements include:

Production Deployments

Staging Deployments

Test Environments

Disaster Recovery Environments

Edge Devices

Embedded Systems

Cloud Instances

Container Images

Retained Artifacts

Deployment removal from one environment must not be treated as complete withdrawal from every environment.

Version-Specific Retirement

Retirement closure may apply to a defined software version rather than an entire product.

RCCLM must preserve:

Retired Version

Continuing Versions

Affected Environments

Shared Dependencies

Version-Specific Support Conditions

Withdrawal Evidence

Remaining Obligations

A version-specific closure must not imply retirement of other active versions.

Partial Retirement Closure

RCCLM recognizes legitimate partial retirement.

Potential conditions include:

One Environment Retired

One Version Retired

One Interface Withdrawn

One Service Capability Discontinued

One Region Transitioned

One User Group Migrated

Partial closure must explicitly identify what has been completed and what remains active.

Support Status Reconciliation

RCCLM must reconcile relevant support conditions established through ESMM.

Relevant elements include:

Support Provider

Support Scope

End-of-Support Status

Maintenance Status

Security Update Availability

Continuing Support Obligations

Unsupported Operational Use

Support Transition Evidence

Support termination must not substitute for operational withdrawal verification.

Migration Outcome Reconciliation

RCCLM must evaluate migration outcomes established through SMM.

Relevant elements include:

Source Software

Destination Software

Migration Scope

Verified Functional Continuity

Verified Data Continuity

User Transition

Interface Transition

Unresolved Migration Obligations

Migration Closure Status

A closed migration case does not automatically establish that source software retirement is complete.

Destination Independence Assessment

Where retirement depends on replacement software, RCCLM must assess whether required operations remain dependent on the source.

Relevant elements include:

Destination Operational Capability

Source Dependency

Fallback Reliance

Historical Access Requirements

Unmigrated Functions

External Consumer Dependencies

Verification Evidence

Destination availability must not be mistaken for independence from the retiring software.

Residual Dependency Reconciliation

RCCLM must reconcile residual dependencies established through RDMM.

Relevant elements include:

Dependency Identity

Dependency State

Required Disposition

Verified Disposition

Responsible Function

Residual Risk

Continuing Obligations

Dependency Closure Status

Every material residual dependency must have a legitimate and traceable disposition.

Residual Dependency Completeness

RCCLM must assess whether residual dependency evaluation is sufficiently complete.

Relevant elements include:

Known Dependencies Covered

Direct Dependencies Assessed

Indirect Dependencies Assessed

External Consumers Assessed

Scheduled Activities Assessed

Data Relationships Assessed

Authority Relationships Assessed

Recovery Dependencies Assessed

Material Uncertainty Preserved

Completeness must not be inferred solely from the absence of newly discovered dependencies.

Data Continuity Reconciliation

RCCLM must assess continuing data obligations.

Relevant elements include:

Data Accessibility

Historical Records

Migration Verification

Data Integrity

Data Retention

Archival Conditions

Recovery Requirements

Data Ownership

Data Disposal Status

Software retirement closure must not silently eliminate continuing data obligations.

Data Retention After Retirement

Some data may legitimately remain after software retirement.

Potential conditions include:

Operational Data Migrated

Historical Data Archived

Evidence Retained

Recovery Data Retained

Data Disposal Pending

Data Retention Obligation Active

RCCLM must distinguish legitimate data retention from unresolved software dependence.

Interface Withdrawal Verification

RCCLM must assess whether interfaces included in the retirement scope have been withdrawn or transferred.

Relevant elements include:

Interface Identity

Affected Consumers

Current Availability

Replacement Interface

External Requests

Authentication State

Routing Configuration

Withdrawal Evidence

A disabled endpoint may remain materially relevant if consumers continue relying on it.

External Consumer Reconciliation

RCCLM must identify unresolved external reliance.

Relevant elements include:

Consumer Identity

Consumed Capability

Migration Status

Notification Evidence

Observed Requests

Alternative Access

Continuing Obligation

Responsible Function

Unknown or unverified consumers may prevent an unqualified closure determination.

Access and Credential Reconciliation

RCCLM must assess access relationships associated with the retiring software.

Potential objects include:

Service Accounts

Application Credentials

API Keys

Certificates

Authentication Tokens

Administrative Permissions

External Access Rights

Shared Credentials

The existence of unused credentials must not automatically be treated as harmless.

Authority Termination Verification

RCCLM must determine whether authority relationships intended to terminate have actually been terminated.

Relevant elements include:

Actor or System Identity

Original Authority

Delegated Permissions

Authorized Actions

External System Access

Revocation Requirements

Revocation Evidence

Remaining Authority

Technical deactivation does not independently establish termination of delegated authority.

Autonomous Agent Retirement Verification

Where autonomous software is retired, RCCLM must assess:

Agent Identity

Execution State

Active Objectives

Scheduled Tasks

Tool Permissions

External Credentials

Persistent Memory References

Pending Actions

Successor Responsibilities

Human Override Conditions

An agent cannot be considered fully reconciled merely because its primary

execution process has stopped.

Automated Execution Path Verification

RCCLM must identify execution paths capable of invoking retired software or related activities.

Potential paths include:

Scheduled Jobs

External Triggers

Event-Driven Workflows

CI/CD Pipelines

Orchestration Systems

API Invocations

Background Workers

Recovery Automation

Agent Tool Chains

Material execution paths must be verified according to the intended retirement scope.

Recovery Dependency Reconciliation

RCCLM must evaluate recovery-related obligations.

Relevant elements include:

Backup Artifacts

Recovery Images

Legacy Restore Tools

Archived Configurations

Recovery Credentials

Recovery Procedures

Historical Dependencies

Retention Requirements

Recovery-related retention may be legitimate even after operational retirement.

Historical Obligation Reconciliation

Some obligations survive retirement.

Potential obligations include:

Audit Records

Incident Evidence

Historical Reporting

Data Retention

Contractual Responsibilities

Regulatory Requirements

Warranty-Related Records

Operational Reconstruction

Legal Preservation Requirements

RCCLM identifies the continuing governance disposition without

independently determining legal enforceability.

Continuing Responsibility

RCCLM must establish responsibility for material obligations that survive closure.

Relevant elements include:

Obligation Identity

Responsible Function

Required Capability

Applicable Authority

Continuing Conditions

Review Requirements

Evidence Location

Termination Conditions

Retirement closure must not create an unassigned responsibility gap.

Retirement Completeness Assessment

RCCLM must evaluate completeness against the approved retirement scope.

Relevant dimensions include:

Software Withdrawal

Migration

Support Transition

Dependency Disposition

Data Continuity

Interface Withdrawal

Access and Authority

Operational Responsibility

Evidence Sufficiency

Residual Obligations

Completeness must remain scope-specific and evidence-supported.

Retirement Evidence Sufficiency

RCCLM must assess whether available evidence is sufficient for the proposed closure disposition.

Relevant considerations include:

Evidence Relevance

Software Identity Match

Environment Coverage

Version Coverage

Temporal Applicability

Verification Method

Operational Observation

Known Limitations

Conflicting Evidence

Material Evidence Gaps

Evidence sufficiency must be assessed against the actual retirement requirements.

Evidence Applicability

Retirement evidence must remain associated with:

Software Object

Version

Configuration

Deployment Environment

Retirement Scope

Execution Time

Verification Time

Responsible Actor

Applicable Authority

Observed Outcome

Evidence from one environment must not automatically establish withdrawal from another.

Retirement Verification Methods

Verification methods may include:

Deployment Inventory Reconciliation

Runtime Monitoring

Execution Path Assessment

Configuration Verification

Interface Activity Assessment

Credential Revocation Verification

Dependency Resolution Verification

Migration Outcome Verification

Operational Owner Confirmation

Independent Technical Review

No single verification method is universally sufficient.

The method must be appropriate to the governed software and retirement scope.

Retirement Deviation Assessment

A retirement deviation occurs when the actual outcome differs materially from the intended retirement condition.

Potential deviations include:

Software Still Executing

Unmigrated Users

Unresolved Dependencies

Active Credentials

Unexpected Interface Activity

Incomplete Data Transfer

Missing Verification Evidence

Unassigned Continuing Obligation

Unauthorized Withdrawal Activity

Material deviations must remain visible in the closure determination.

Retirement Exception Assessment

Some retirement conditions may require an authorized exception.

Relevant elements include:

Exception ID

Affected Requirement

Reason

Applicable Authority

Alternative Control

Residual Risk

Validity Conditions

Review Requirements

Evidence

An exception cannot convert an unverified retirement outcome into a

verified one.

Retirement Uncertainty

RCCLM must preserve uncertainty that affects closure eligibility.

Potential uncertainty includes:

Unknown Software Instance

Unverified Execution Path

Unknown External Consumer

Incomplete Dependency Discovery

Unverified Data Access

Unclear Credential State

Conflicting Migration Evidence

Unknown Recovery Requirement

Material uncertainty must not be concealed by a completed project status.

Closure Eligibility

A retirement case is eligible for a proposed closure disposition only when the applicable closure criteria have been assessed.

Relevant criteria include:

Retirement Scope Established

Intended State Defined

Actual State Assessed

Material Withdrawal Verified

Migration Outcomes Reconciled

Residual Dependencies Addressed

Data Obligations Addressed

Authority Conditions Reconciled

Continuing Responsibilities Assigned

Evidence Sufficient for the Proposed Disposition

Eligibility must remain distinct from formal closure authorization.

Closure Preconditions

Material closure decisions may require:

Applicable Retirement Decision

Verified Software Identity

Verified Retirement Scope

Withdrawal Evidence

Migration Reconciliation

Dependency Reconciliation

Data Continuity Assessment

Authority Reconciliation

Residual Obligation Disposition

Competent Closure Authority

Unmet mandatory preconditions must prevent an unqualified completed-retirement determination.

Closure Authority

RCCLM must identify the authority competent to establish the applicable retirement closure disposition.

Relevant elements include:

Software Ownership

Operational Responsibility

Retirement Decision Authority

Closure Approval Authority

Data Governance Authority

Security Authority

External Approval Requirements

Delegation Conditions

Authority to execute retirement activities does not automatically establish authority to approve final closure.

Closure Responsibility

RCCLM must identify responsibility for:

Closure Assessment

Evidence Collection

Operational Verification

Dependency Reconciliation

Data Obligation Reconciliation

Authority Verification

Closure Recommendation

Closure Decision

Continuing Obligation Monitoring

Responsibility assignment must remain aligned with applicable capability and authority.

Closure Decision

The closure decision must establish the justified disposition of the retirement case.

Relevant elements include:

Closure Case Identity

Retirement Scope

Intended Retirement State

Verified Actual State

Evidence Basis

Unresolved Conditions

Residual Obligations

Applicable Authority

Closure Classification

Effective Time

The decision must not assert a stronger retirement state than the evidence supports.

Closure Classification

RCCLM recognizes the following governance classifications:

Closure Assessment Pending

Verification Active

Closure Ready

Closure Conditionally Ready

Closure Blocked

Retirement Verified — Complete

Retirement Verified — Partial Scope

Retirement Closed With Governed Residual Obligations

Conditional Closure

Closure Deferred

Closure Rejected

Closure Invalidated

Reassessment Required

Retirement Case Reopened

Unknown

Conflicted

Each classification must preserve its scope, supporting evidence, applicable authority, and conditions.

Verified Complete Retirement

A retirement case may be classified as verified complete when:

• The defined retirement scope is established

• The intended operational withdrawal is verified

• Required migration outcomes are reconciled

• Material residual dependencies have legitimate dispositions

• Applicable authority relationships have been reconciled

• Continuing data and historical obligations are governed

• Evidence is sufficient for the closure determination

• Competent closure authority has established the final disposition

Verified complete retirement does not mean that all historical records or obligations have ceased to exist.

Verified Partial Retirement

Partial retirement may be verified when a defined subset of the intended retirement scope has been completed.

The closure record must preserve:

Completed Scope

Remaining Scope

Verified Withdrawal Conditions

Continuing Software Instances

Outstanding Dependencies

Responsible Functions

Remaining Retirement Requirements

Partial closure must not be represented as complete retirement.

Conditional Closure

Conditional closure may be appropriate where a limited closure disposition is legitimate but specific continuing conditions remain.

Relevant elements include:

Closure Scope

Conditions

Responsible Functions

Residual Obligations

Review Requirements

Validity Conditions

Reopening Triggers

Applicable Authority

Conditional closure must not be used to bypass mandatory withdrawal verification or unresolved critical conditions.

Closure With Governed Residual Obligations

A retirement case may be legitimately closed while continuing obligations remain.

Potential obligations include:

Historical Data Retention

Audit Evidence Preservation

Archived Software Artifacts

Recovery Documentation

Contractual Responsibilities

External Reporting

Incident Cooperation

Restricted Historical Access

Such closure requires identifiable responsibility, applicable authority,

and continuing governance.

Closure Deferral

Closure must be deferred where material requirements remain outstanding.

Potential reasons include:

Withdrawal Not Verified

Migration Incomplete

Critical Dependency Unresolved

Authority State Unclear

Data Continuity Unverified

Insufficient Evidence

Material Operational Risk

Unknown Execution Path

Deferral must preserve the outstanding conditions and responsible functions.

Closure Rejection

Closure may be rejected when the proposed retirement disposition is not supported.

Potential reasons include:

Software Remains Operational Contrary to Intended Scope

Material Withdrawal Failed

Unauthorized Retirement Activity

Critical Continuity Requirement Not Satisfied

Unresolved Material Dependency

Materially Conflicting Evidence

Closure Authority Absent

Rejection must remain distinguishable from temporary deferral.

Administrative Closure Boundary

Administrative project completion may occur independently of verified retirement closure.

Potential administrative events include:

Project Budget Closed

Project Team Released

Change Ticket Completed

Migration Project Completed

Vendor Contract Ended

Inventory Status Updated

Retirement Announcement Published

None of these events independently establishes that the intended software withdrawal has occurred.

Operational Finality

Operational finality concerns the established termination of the software's

intended operational role within the defined retirement scope.

Relevant elements include:

Execution State

Deployment State

Interface State

User Reliance

Dependency State

Authority State

Operational Responsibility

Verification Evidence

Operational finality is scope-specific and does not eliminate legitimate historical obligations.

Retirement Finality Boundary

RCCLM must distinguish:

Operational Finality

Support Finality

Migration Finality

Dependency Disposition Finality

Administrative Finality

Historical Obligation Continuity

These conditions may occur at different times.

No single condition automatically establishes every other condition.

Post-Closure Monitoring

Some retirement cases require continued observation after closure.

Potential monitoring concerns include:

Unexpected Software Reactivation

Legacy Interface Requests

Unauthorized Credential Use

Residual Dependency Reactivation

Data Access Failure

Recovery Obligation Changes

External Consumer Activity

Continuing Obligation Expiration

Monitoring must remain linked to an identifiable responsible function and justified scope.

Closure Reassessment

RCCLM must support reassessment when material evidence or conditions change.

Potential triggers include:

Previously Unknown Software Instance Discovered

Unexpected Execution Detected

Residual Dependency Reactivated

Data Continuity Failure

Authority Revocation Failure

Migration Outcome Invalidated

Closure Evidence Contradicted

Continuing Obligation Changed

Reassessment must preserve the original closure decision and its evidence.

Closure Invalidation

A closure determination may be invalidated when material evidence establishes that its required conditions were not satisfied.

Relevant elements include:

Original Closure Decision

Invalidating Evidence

Affected Scope

Material Discrepancy

Applicable Authority

Required Corrective Action

Revised Governance Status

Invalidation must not silently overwrite the historical closure record.

Retirement Case Reopening

A retirement case may be reopened when further governance action is required.

Potential triggers include:

Unexpected Software Operation

Unresolved Critical Dependency

Unauthorized Continuing Access

Incomplete Migration Discovered

Historical Data Inaccessibility

New Material Evidence

Invalid Closure Determination

Reopening must preserve continuity with the original retirement case.

Closure Record Integrity

RCCLM must preserve the integrity of the closure record.

Relevant elements include:

Closure Case Identity

Software Identity

Retirement Scope

Decision Authority

Decision Time

Evidence References

Closure Classification

Residual Obligations

Record Version

Reassessment History

Material revisions must remain traceable.

Closure Evidence Continuity

Evidence must remain accessible for the applicable governance period.

Potential evidence includes:

Retirement Plans

Support Transition Records

Migration Profiles

Dependency Disposition Records

Withdrawal Verification

Authority Revocation Evidence

Operational Monitoring

Closure Decisions

Reopening Records

Evidence retention must follow applicable obligations and specialized governance requirements.

Closure and Operational Reality

RCCLM must distinguish declared retirement states from actual operational conditions.

A software inventory may show an application as retired while a recovery environment continues executing it.

A cloud service may be marked deleted while external consumers continue accessing a retained endpoint.

A migration project may be closed while historical data remains inaccessible.

An autonomous agent may be disabled while external credentials remain valid.

Retirement closure must therefore follow verified operational reality.

Declared Retirement State ≠ Verified Operational Retirement State.

Governance Requirements

A conforming RCCLM implementation must satisfy the following requirements. Retirement Identity and Scope

• Establish a stable retirement closure case identity

• Identify the retiring software object

• Establish the authorized retirement scope

• Identify applicable software versions and environments

• Preserve the intended retirement state

• Establish the actual software state

• Preserve relevant retirement execution evidence

• Identify excluded and continuing software objects

Withdrawal and Transition Verification

• Verify material software withdrawal conditions

• Identify remaining execution paths

• Reconcile deployment states

• Reconcile support conditions

• Reconcile migration outcomes

• Reconcile source–destination dependencies

• Reconcile interface withdrawal

• Reconcile data continuity

• Reconcile access and authority conditions

• Reconcile automated and autonomous execution where applicable

Residual Obligations

• Identify material residual dependencies

• Verify their applicable dispositions

• Identify continuing data obligations

• Identify historical and recovery obligations

• Establish responsibility for continuing obligations

• Preserve applicable authority

• Identify residual risk

• Preserve material uncertainty

• Identify post-closure monitoring requirements

• Preserve unresolved conditions

Closure Eligibility and Decision

• Establish closure criteria

• Evaluate evidence sufficiency

• Identify retirement deviations

• Establish closure eligibility

• Identify competent closure authority

• Preserve closure decision evidence

• Classify the resulting retirement disposition

• Distinguish partial, conditional, and complete retirement

• Prevent administrative completion from substituting for operational

• Prevent migration completion from substituting for retirement

verification. closure.

Continuity and Reassessment

• Preserve closure record integrity

• Preserve historical evidence

• Establish reassessment triggers

• Support closure invalidation

• Support retirement case reopening

• Preserve continuity between original and revised decisions

• Maintain responsibility for retained obligations

• Prevent retirement closure from being treated as elimination of

historical responsibility.

Governance Logic

Retirement Case → Closure Case Identity → Authorized Retirement Scope →

Intended Retirement State → Actual Software State → Withdrawal Evidence → Support Reconciliation → Migration Reconciliation → Residual Dependency Reconciliation → Data Continuity → Interface State → Authority and Access State → Continuing Obligations → Evidence Sufficiency → Retirement Completeness → Closure Eligibility → Competent Authority → Closure Decision → Software Retirement Closure Profile

Where retirement is fully verified:

Intended Withdrawal → Verified Actual Withdrawal → Residual Obligations

Governed → Closure Authorized → Verified Complete Retirement

Where only part of the scope is verified:

Defined Retirement Scope → Verified Subset → Remaining Scope Identified →

Partial Closure

Where legitimate obligations continue:

Operational Withdrawal Verified → Continuing Obligations Identified →

Responsibility and Authority Established → Governed Retention → Closure With Residual Obligations

Where material evidence is insufficient:

Closure Request → Evidence Gap → Verification Pending / Closure Deferred

Where the actual state contradicts the intended state:

Closure Assessment → Material Deviation → Closure Blocked / Rejected →

Corrective Governance

Where new evidence invalidates closure:

Existing Closure → New Material Evidence → Reassessment → Invalidation

Where Required → Reopened Retirement Case

RCCLM must not equate administrative completion with verified retirement finality.

Governance Outputs

The principal output is:

Software Retirement Closure Profile

Supporting outputs may include:

Retirement Closure Case Identity

Retirement Scope Verification Record

Intended–Actual Retirement State Assessment

Software Withdrawal Verification Record

Deployment State Reconciliation

Support Status Reconciliation

Migration Outcome Reconciliation

Residual Dependency Reconciliation

Data Continuity Assessment

Interface Withdrawal Assessment

Authority and Access Reconciliation

Autonomous Execution Reconciliation

Retirement Evidence Sufficiency Assessment

Retirement Completeness Profile

Closure Eligibility Record

Closure Authority Record

Retirement Closure Decision

Continuing Obligation Register

Post-Closure Monitoring Requirements

Closure Reassessment Record

Closure Invalidation Record

Retirement Reopening Record

Minimum Implementation Framework

A minimum RCCLM implementation must establish eight governance layers. Layer 1 — Closure Identity and Scope

Identify the retirement case, software object, authorized retirement scope,

and intended retirement state. Layer 2 — Actual State Verification

Establish the actual execution, deployment, interface, access, and

operational conditions of the retiring software. Layer 3 — Transition Reconciliation

Reconcile support termination, migration outcomes, and other retirement

activities with the intended state. Layer 4 — Residual Dependency and Obligation Reconciliation

Identify and verify the dispositions of material dependencies, data

relationships, authority conditions, and continuing responsibilities. Layer 5 — Evidence and Completeness Assessment

Evaluate evidence sufficiency, retirement completeness, deviations, and

material uncertainty. Layer 6 — Closure Eligibility and Authority

Establish the applicable closure criteria and competent authority for the

proposed disposition. Layer 7 — Closure Decision

Record a justified classification of complete, partial, conditional,

deferred, rejected, or otherwise qualified retirement closure. Layer 8 — Continuity and Reassessment

Preserve closure evidence, continuing obligations, post-closure monitoring,

invalidation conditions, and reopening capability.

The framework may be implemented through software asset inventories, deployment monitoring, configuration systems, migration platforms, dependency registers, service management tools, or AI-assisted verification.

The implementation technology must adapt to the retirement governance requirements.

Cross-Module Boundary

RPM → RCCLM

RPM establishes the intended retirement state, scope, feasibility,

transition requirements, and initial readiness criteria.

RCCLM verifies whether the resulting retirement outcome satisfies those

conditions.

ESMM → RCCLM

ESMM establishes the resulting support status and continuing support

obligations.

RCCLM reconciles these conditions with retirement closure requirements.

SMM → RCCLM

SMM establishes migration outcomes, verification evidence, and unresolved

migration obligations.

RCCLM evaluates whether those outcomes support legitimate retirement

closure.

RDMM → RCCLM

RDMM establishes residual dependency dispositions, verification evidence,

and continuing responsibilities.

RCCLM evaluates whether those dispositions satisfy the applicable

retirement closure criteria.

RCCLM establishes the retirement closure disposition. It does not replace the specialized governance processes that produce the underlying evidence.

Cross-Architecture Boundary

RCCLM governs retirement closure within SRTS.

It may depend on evidence and conditions governed by other SLGA™ Parent Standards.

SRS — Software Requirements Standard

Requirements relevant to replacement capabilities and continuing software

obligations.

SCS — Software Creation Standard

Creation records for replacement or supporting software.

SDS — Software Dependency Standard

Dependency identity, suitability, continuity, and risk.

SCFS — Software Configuration Standard

Artifact, version, configuration, and deployment identity.

SQS — Software Quality Standard

Quality verification of replacement or modified software.

SSS — Software Security Standard

Security requirements, vulnerability conditions, and remediation evidence.

SRLS — Software Release Standard

Release readiness and authorization of successor software.

SDPS — Software Deployment Standard

Deployment state, operational transition, and rollback or recovery.

SCMS — Software Change Management Standard

Change closure and lifecycle re-entry associated with retirement activities.

SLCS — Software Lifecycle Control Standard

Lifecycle state consistency, transition control, exception governance, and

continuity.

RCCLM may also interact with applicable OOF® identity, authority, accountability, operational reality, data, autonomous systems, and physical-system governance mechanisms.

It does not replace their specialized governance functions.

Practical Applications

Enterprise Application Decommissioning

Verifying withdrawal of obsolete applications and reconciliation of

historical data, user access, and residual integrations.

Cloud Service Retirement

Establishing whether cloud services have been withdrawn from the intended

environments and whether external consumers and credentials remain active.

Software Supply-Chain Modernization

Verifying retirement of unsupported libraries, runtimes, and software

components.

AI Service Retirement

Reconciling model endpoints, dependent applications, data obligations, and

continuing operational access.

Autonomous Agent Decommissioning

Verifying termination of execution paths, delegated permissions, scheduled

activities, and unresolved responsibilities.

Industrial and Embedded Software

Assessing whether retired software has been withdrawn without leaving

unsafe or unsupported physical-system dependencies.

Critical Infrastructure

Establishing evidence-supported retirement closure while preserving

recovery, auditability, and continuing operational obligations.

Practical Use Case 1 — Enterprise Application Retirement Closure

An organization completes a project to replace a legacy enterprise

application.

The replacement system is operational, users have migrated, and the original application has been removed from the primary production environment.

The project team submits the retirement case for closure.

RCCLM evaluates:

Retirement Scope → Intended State → Actual State → Migration Outcome →

Residual Dependencies → Data Continuity → Withdrawal Verification → Closure Eligibility

The assessment identifies:

• The primary production application is no longer executing

• Required business functions have been verified on the replacement

• Historical financial records remain accessible through a governed

• A disaster recovery environment still contains an executable copy of

• The recovery copy has a legitimate retention purpose but requires

• The original retirement plan did not require immediate destruction of

system. archive. the original application. explicit access restrictions and responsibility assignment. all recovery artifacts.

RCCLM distinguishes the retired operational production capability from the

retained recovery artifact.

The resulting disposition may be:

Retirement Verified — Complete for Defined Production Scope; Recovery

Artifact Retained Under Governance.

The closure record preserves the recovery obligation without incorrectly

treating it as evidence of continuing production operation.

Operational Retirement ≠ Elimination of Every Historical Artifact.

Practical Use Case 2 — Autonomous Agent Retirement Closure

An organization retires an autonomous software engineering agent and

replaces it with a successor.

The original agent's execution environment has been stopped, and its

assigned development tasks have been transferred.

A closure request states that retirement is complete.

RCCLM evaluates:

Agent Identity → Execution State → Scheduled Activities → Tool Permissions

→ External Credentials → Task Transfer → Residual Dependencies → Closure Evidence

The assessment identifies:

• The original agent's primary execution process has stopped

• The successor agent has independently authorized access

• Task continuity has been verified

• One external automation workflow can still invoke actions using the

• The original credentials remain valid

• Historical task records are retained for audit purposes

original agent's credentials.

RCCLM determines that the intended authority termination has not been verified.

The retirement case is classified:

Closure Blocked — Residual Execution and Authority Conditions Outstanding.

After the external workflow is disabled, the original credentials are revoked, and the resulting conditions are verified, the case may be reassessed.

Agent Deactivated ≠ Agent Retirement Verified.

Critical Distinctions

Retirement Announced ≠ Retirement Executed.

Retirement Executed ≠ Retirement Verified.

Software Deleted ≠ Operational Withdrawal Verified.

Software Stopped ≠ Every Execution Path Terminated.

Deployment Removed ≠ Every Environment Reconciled.

Migration Completed ≠ Retirement Completed.

Destination Operational ≠ Source Independent.

Support Terminated ≠ Software Withdrawn.

Dependency Disposition Recorded ≠ Dependency Disposition Verified.

Residual Dependency Retained ≠ Residual Dependency Ungoverned.

Data Archived ≠ Data Accessibility Verified.

Interface Disabled ≠ External Consumer Transition Verified.

Credential Unused ≠ Credential Revoked.

Agent Execution Stopped ≠ Agent Authority Terminated.

Administrative Closure ≠ Operational Finality.

Partial Retirement ≠ Complete Retirement.

Closure Eligible ≠ Closure Authorized.

Conditional Closure ≠ Unqualified Completion.

Retirement Closure ≠ Elimination of Historical Responsibility.

Previously Verified Closure ≠ Permanently Irreversible Closure

Determination.

Canonical Boundary

Retirement Closure Control Module (RCCLM) governs the evidence-supported

verification, reconciliation, authorization, classification, and continuing validity of software retirement closure.

It establishes the relationship between the intended retirement state and

the actual operational condition of the software.

It reconciles withdrawal evidence, support transitions, migration outcomes,

residual dependencies, data relationships, access permissions, authority conditions, and continuing obligations.

It permits legitimate partial or qualified closure where the scope and conditions are explicitly established.

It does not permit an unqualified retirement determination where mandatory withdrawal conditions remain unverified.

RCCLM does not replace the specialized processes responsible for executing software withdrawal, migration, security remediation, dependency resolution, or data management.

Its principal governance output is the:

Software Retirement Closure Profile

This output establishes what was intended to retire, what actually ceased

operating, what was verified, what remains governed, and why the resulting retirement disposition is legitimate.

Canonical Closing Statement

Software retirement is not complete because an organization has finished the work it planned. It is complete only when the required retirement outcome has been established.

A project can close while software remains deployed.

A replacement can operate while the original application remains necessary.

A cloud service can be deactivated while external integrations continue attempting to use it.

A legacy system can disappear from production while essential recovery dependencies remain.

An autonomous agent can stop executing while its permissions and external automation paths remain active.

And historical data, evidence, and responsibilities may legitimately survive the software's operational withdrawal.

RCCLM establishes the governance mechanism required to distinguish these

conditions.

It connects the intended retirement state to the actual operational state.

It requires withdrawal to be verified rather than declared.

It reconciles migration, support, dependencies, data, and authority.

It prevents unresolved obligations from disappearing behind administrative

completion.

It permits legitimate continuing relationships without confusing them with

active software operation.

It establishes a traceable closure decision supported by evidence and

competent authority.

And it preserves the ability to reassess retirement closure when material new evidence changes the established understanding of operational reality.

Retirement finality does not mean that nothing remains. It means that the software's intended operational role has ended and everything that legitimately remains has an identifiable governance disposition.

Retirement Closure Control Module (RCCLM) transforms software retirement

closure from an administrative endpoint into a verifiable governance determination—establishing when software has truly completed its intended operational lifecycle and ensuring that its remaining obligations do not disappear with it.